

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EE)

Nov/Dec 2012

EE-217 ENGINEERING ELECTROMAGNETICS

Date: 30.11.2012, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Define following: [03]

- Coulomb's Law
- Divergence Theorem
- Gauss Law

(b) Transform the following vectors to cylindrical co-ordinates at the point specified [02]

(1) $5\mathbf{a}_x$ at P ($\rho=4$, $\phi=120^\circ$, $z=-1$)

(c) Transform the following vectors to spherical co-ordinates at the point specified [02]

(1) $5\mathbf{a}_x$ at P ($r=4$, $\theta=25^\circ$, $\phi=120^\circ$)

Q - 2 (a) Derive the Equation (Expression) of Electric Field Intensity (E) at any point due to Uniform Line charge distribution with density. [07]

(b) If $\vec{A} = 2\mathbf{a}_x + 6\mathbf{a}_y - 3\mathbf{a}_z$ & $\vec{B} = -3\mathbf{a}_x - 4\mathbf{a}_y - 5\mathbf{a}_z$ find out [05]

- I. a unit vector in the direction of $\vec{A} - \vec{B}$
- II. the magnitude of $\vec{A} + \vec{B}$.
- III. $[(\vec{A} + 3\vec{B}) / |(\vec{A} - 3\vec{B})|] \cdot \vec{B}$.
- IV. Find Area of parallelogram made by the above two vectors.
- V. Find the Area of Triangle for the same

(c) Select the appropriate choice for given question, write the (option & answer) in the answer booklet. [02]

1. Direction of E due to Point charge, Line charge & Sheet charge respectively are

- | | | |
|-------------------------|---------------------|----------------|
| a) Radially out, | Radially out or in, | & Normal |
| b) Spherical out or in, | Radially out or in, | & Normal |
| c) Spherical out, | Radially in, | & Spherical in |
| d) Radially in | Radially out | & Normal |

2. Differential Length in Cylindrical co-ordinate system is given by _____

- a.) $dL = \rho d\rho \hat{a}_\rho + \rho d\phi \hat{a}_\phi + dz \hat{a}_z$ b.) $dL = \rho d\rho \hat{a}_\rho + d\phi \hat{a}_\phi + dz \hat{a}_z$
 c.) $dL = \rho d\rho \hat{a}_\rho + \rho d\phi \hat{a}_\phi + dz \hat{a}_z$ d.) $dL = d\rho \hat{a}_\rho + d\phi \hat{a}_\phi + dz \hat{a}_z$

OR

- Q - 2 (a) Derive the Expression of Electric field Intensity (E) due to uniform surface charge distribution have surface charge density ρ_s ? [6]
 (b) Obtain the Expression of Divergence of D for differential volume element. [5]
 (c) Derive the expression of energy or work done in moving a point charge in an external electric field. [3]
- Q - 3 (a) State and Prove Divergence theorem. [6]
 (b) Derive the expression for finding Electric field intensity (E) from potential (V) giving the gradient relation Between E and V. [6]
 (c) 1. Breakdown of Dielectric material (i.e. insulation) in electrical system or electrical Circuits takes place due to increase in which electrical quantity [2]

 2. Unit of Electrical Potential (Voltages) is _____.
 a.) Volts b.) Joules/Coulomb c.) Both b & a d.) Volts/coulomb

OR

- Q - 3 (a) An infinite long, uniform line charge is located at (y=3, z=5) if $\rho_L=30\text{nC/m}$, Find E at [5]
 1.) the origin
 2.) Point Pb (0,6,1)
 3.) Point Pc (5,6,1)
 (b) Calculate resultant E at M (3,-4,2) in free space caused by charges [3]
 1.) Charge $Q_1=2\mu\text{C}$ at point P1(0,0,0,) and charge $Q_2=3\mu\text{C}$ at P2(-1,2,3)
 (c) Explain Dipole and Electrical field of a dipole at any given point. [6]

SECTION – II

Q - 4 (a) Define following: [05]

- 1.) BIO SAVART's LAW
- 2.) Ampere's Circuital law
- 3.) Curl Theorem-Statement
- 4.) Faraday's Law of Electromagnetic Induction
- 5.) Lenz's law

(b) State and prove Stoke's theorem. [02]

Q - 5 (a) Make a comparison between Conductors, Semiconductors & Insulators. [03]

(b) Discuss the continuity equation of current. [04]

(c) Derive the Expression of Curl of $\vec{H}$ or discuss the point form of Ampere's Circuital Law. Also give the expressions for the same in various co-ordinate systems. [07]

OR

Q - 5 (a) Differentiate between the Electric and Magnetic circuits. (Any 7 points) [07]

(b) For the given coaxial square coils, has [07]

N_1 =no of turns =1200, N_2 =no of turns =1200,

Length l_1 = 25 cm, Length l_2 = 25 cm,

coil₁ area= 3x3 cm coil₂area= 1.2x1.2 cm

- 1.) For inner coil and core removed, find with current=1 mA Determine magnetic field intensity (H) with in outer coil.
- 2.) Calculate the self inductance of the outer coil.
- 3.) Determine the mutual inductance between the coil.

Q - 6 (a) Obtain the Expression of Magnetic field intensity $\vec{H}$ at any point due to conductor carrying current I placed on Z axis from $-\infty$ to ∞ . [07]

(b) Discuss the Application of Amperes Circuital Law for finding magnetic field intensity $\vec{H}$ due to at various radii of Coaxial Conductor. Discuss all the cases [07]

OR

Q - 6 (a) Obtain the boundary conditions for magnetic Materials. [5]

(b) Define Current & Current Density, and derive Relationship between Current Density J Volume charge Density (ρ_v) and its Velocity V. [5]

(c) Derive Maxwell's equation from Faraday's law in point form and integral form. [4]
